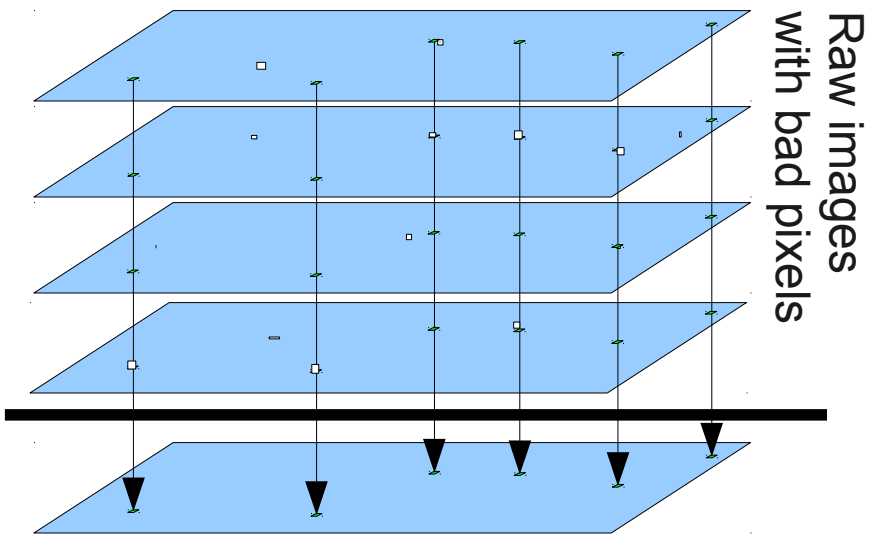
Normalized Median Combination

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Median Combination

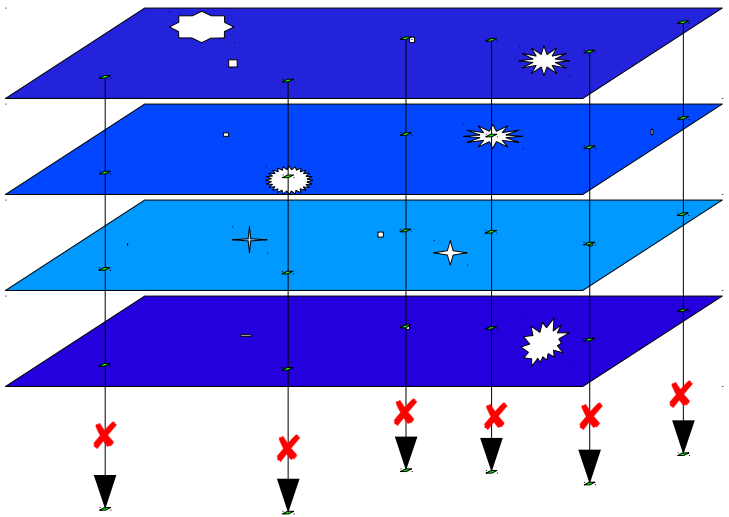


Final image
No bad pixels

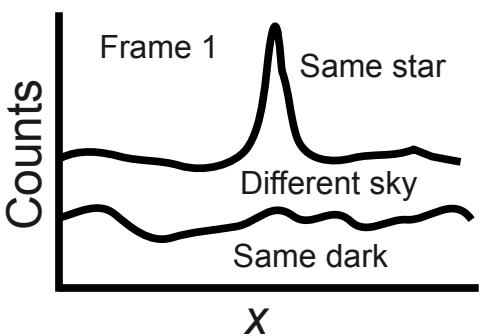
1. Stack the images
2. At each (*x,y*) position take the median
3. Make a new image using those values
4. “Exceptional” values now gone
5. Remaining values near average for this (*x,y*) in stack

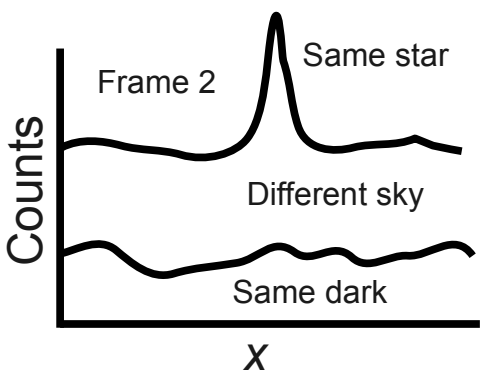
This is good for combining biases, darks, and flats that have transient bad pixels, like cosmic-ray hits.

Normalized Median Combination



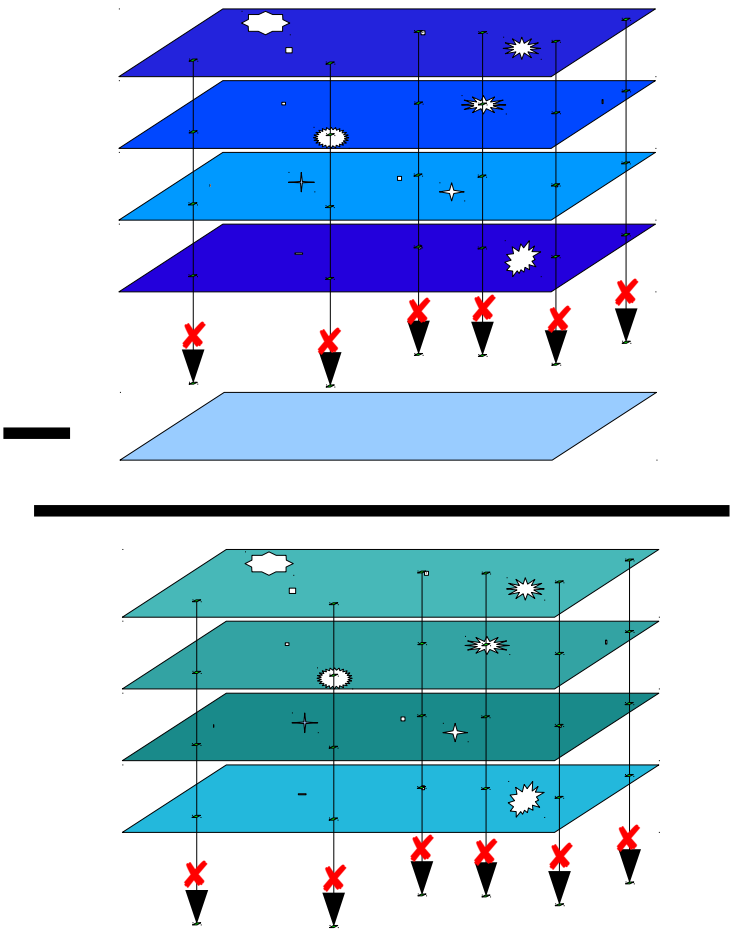
What about target (on-sky) images? If we dither (shift telescope pointing around), we can get sky for free. But, if sky level varies, median makes no sense! Must correct and normalize before taking median.





- Remove dark
- Normalize sky
- Remove sky
- Denormalize stars

Normalized Median Combination

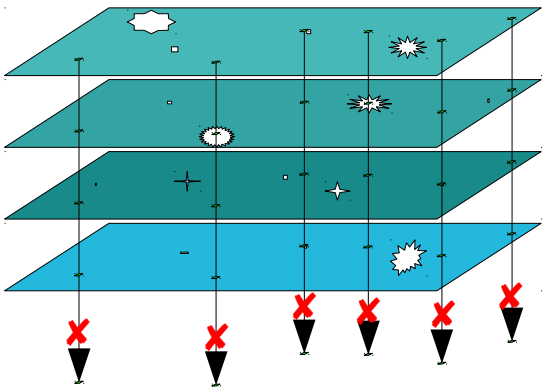


Raw frames with stars, varying sky

Subtract median-combined dark frame

Dark-subtracted frames with varying sky (still can't take median)

Normalized Median Combination

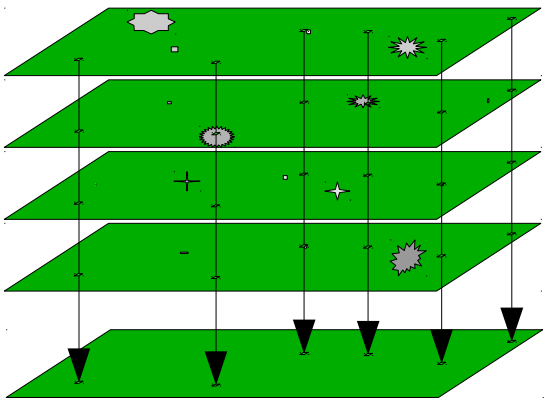


÷ 9000 =

÷ 10000 =

÷ 11000 =

÷ 8000 =



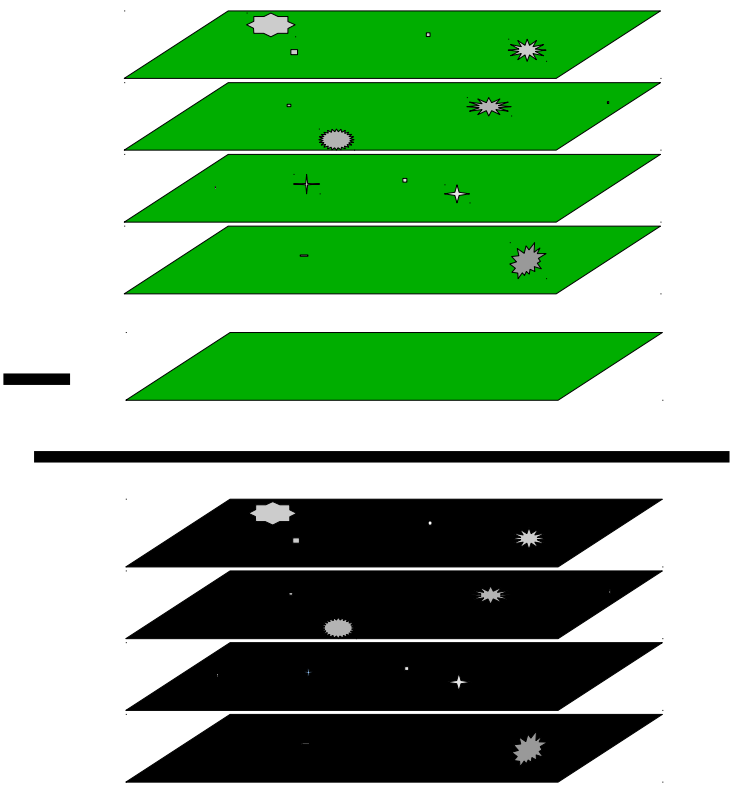
Dark-subtracted frames with varying sky (still can't take median)

Divide by frame median (typical background)

Normalized frames. Typical Background = 1. Stars normalized differently in each frame.

Median-combine to remove stars.

Normalized Median Combination

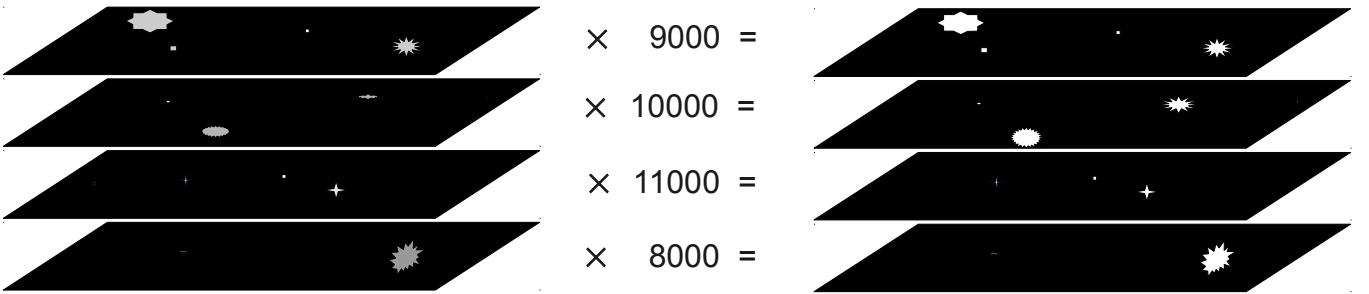


Normalized frames

Subtract normalized, median-combined sky frame

Sky-subtracted frames with stars normalized to original sky in each frame.

Normalized Median Combination



Sky-subtracted frames with stars normalized to original sky in each frame

Multiply by original sky (frame median from before)

Sky- and dark-subtracted frames!